

Label data
training

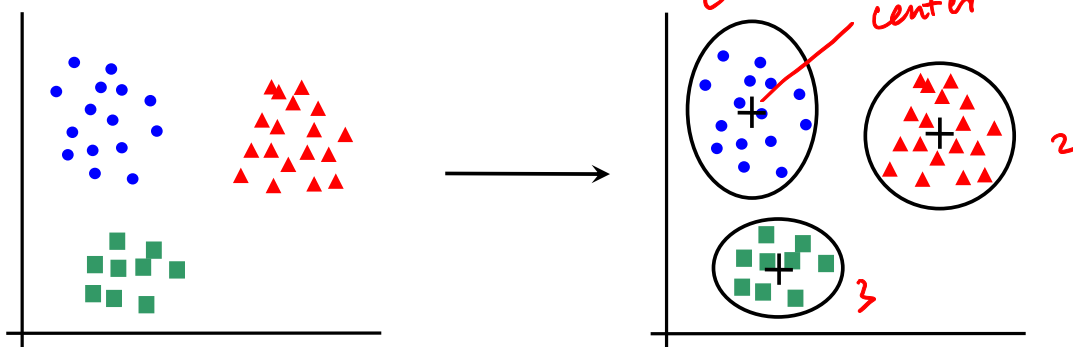
unsupervised learning (cluster)

Lecture 13: Gaussian Mixture Models

Slides from A. Zisserman

- Review: K-means algorithm (hard clustering) data point
- GMM and the EM algorithm (soft clustering) $p(x_i|g)$

- clustering



K-means algorithm

K-means algorithm

you pick $K \geq 2$

Partition data into K sets (*K clusters*)

- Initialize: choose K centres (at random)

Centroid

- Repeat:

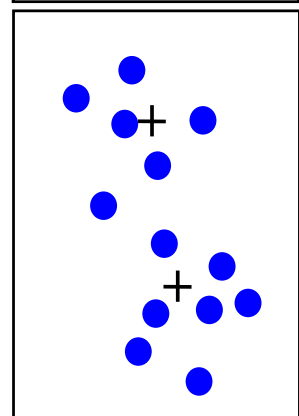
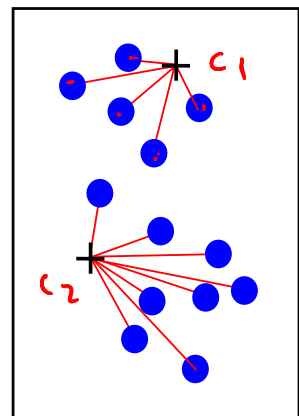
1. Assign points to the nearest centre

*closest
Euclidean distance*

2. New centre = mean of points assigned to it

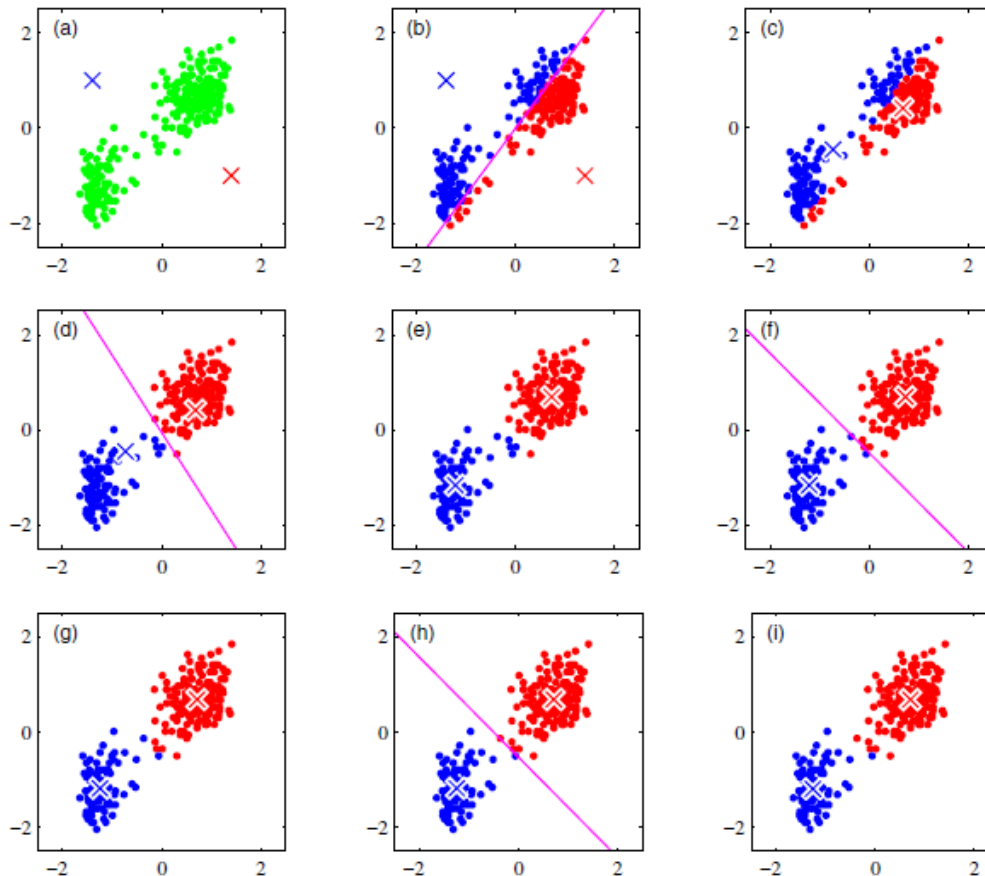
move the centroid

- Until no change



Example

$k=2$
random initial



Converge
↓
center doesn't change
assignment doesn't change

Cost function

minimize distortion

K-means minimizes a measure of distortion for a set of vectors $\{x_i\}, i = 1, \dots, N$

$$D = \sum_{i=1}^N \|x_i^k - c_k\|^2$$

where x_i^k is the subset assigned to the cluster k . The objective is to find the set of centres $\{c_k\}, k = 1, \dots, K$ that minimize the distortion:

$$\min_{c_k} \sum_{i=1}^N \|x_i^k - c_k\|^2$$

Introducing binary assignment variables r_{ik} , the distortion can be written as

$$D = \sum_{i=1}^N \sum_{k=1}^K r_{ik} \|x_i - c_k\|^2$$

where if x_i is assigned to cluster k then

$$r_{ij} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

$r_{ik} = 1$

iterative algorithm

random initial

K-means
binary assignment

x_i
↓
cluster

$c_1 = 1, c_2 = 0$
 $c_1 = 0, c_2 = 1$

Minimizing the Cost function

We want to determine

$$\min_{\mathbf{c}_k, r_{ik}} D = \sum_{i=1}^N \sum_{k=1}^K r_{ik} \|\mathbf{x}_i - \mathbf{c}_k\|^2$$

$\mathbf{c}_k, r_{ik}$

Step 1: minimize over assignments r_{ik}

$\mathbf{c}_k \rightarrow r_{ik}$ (shortest distance)

Each term in $\mathbf{x}_i$ can be minimized independently by assigning $\mathbf{x}_i$ to the closest centre $\mathbf{c}_k$

Step 2: minimize over centres $\mathbf{c}_k$

$$\frac{d}{d\mathbf{c}_k} \sum_{i=1}^N \sum_{k=1}^K r_{ik} \|\mathbf{x}_i - \mathbf{c}_k\|^2 = 2 \sum_{i=1}^N r_{ik} (\mathbf{x}_i - \mathbf{c}_k) = 0$$

local minimum

Hence

$$\mathbf{c}_k = \frac{\sum_{i=1}^N r_{ik} \mathbf{x}_i}{\sum_{i=1}^N r_{ik}}$$

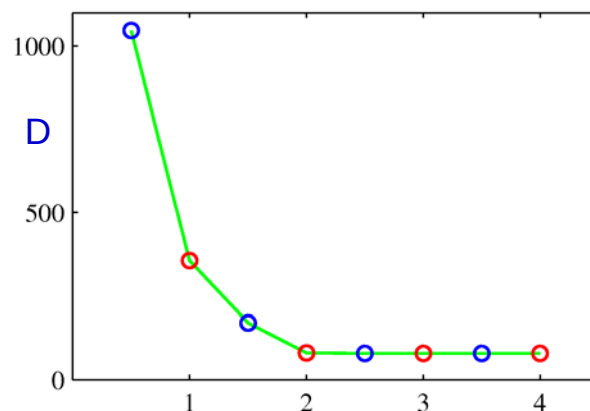
mean

k - mean ++

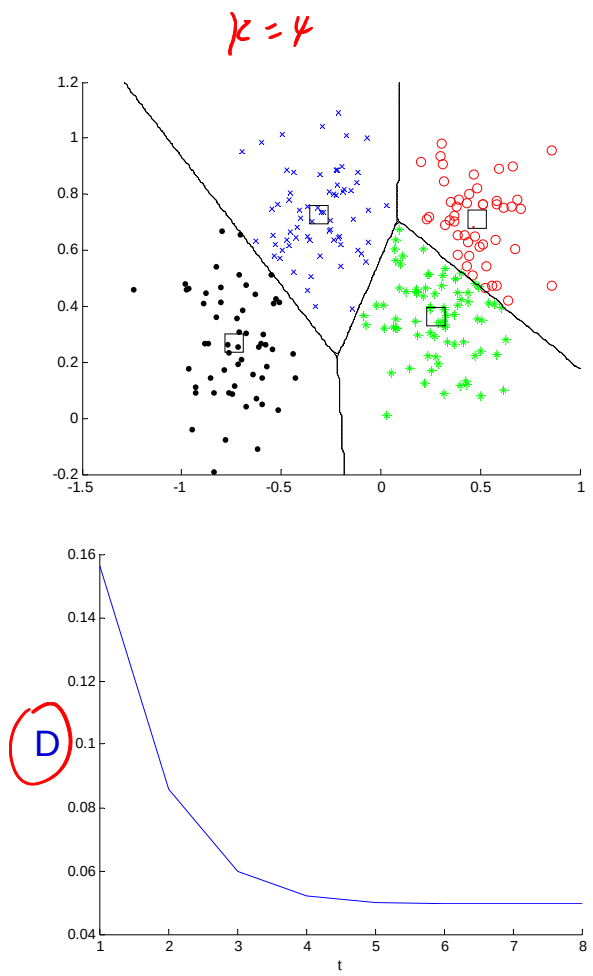
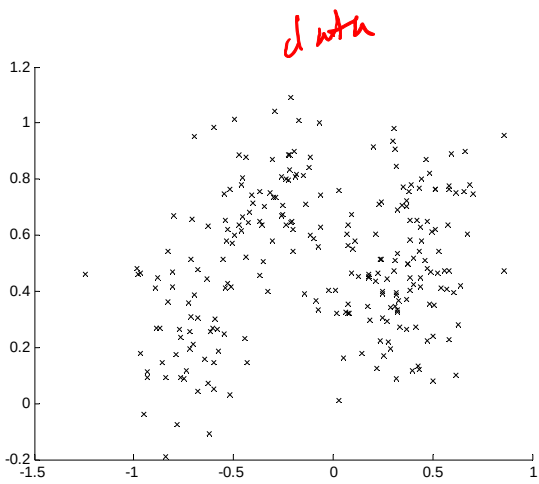
i.e. $\mathbf{c}_k$ is the mean (centroid) of the vectors assigned to it.

Note, since both steps decrease the cost D , the algorithm will converge – but it can converge to a local rather than global minimum.

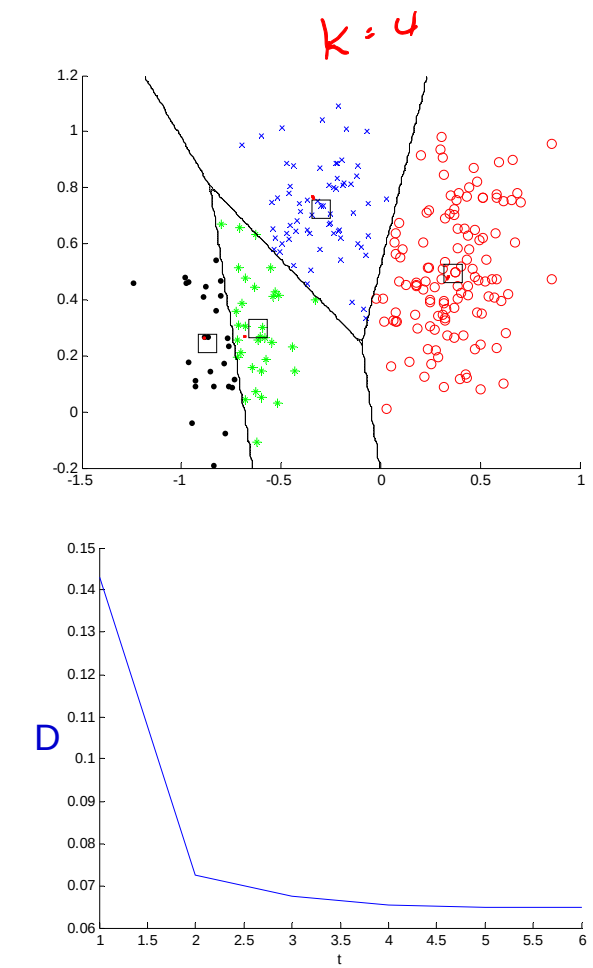
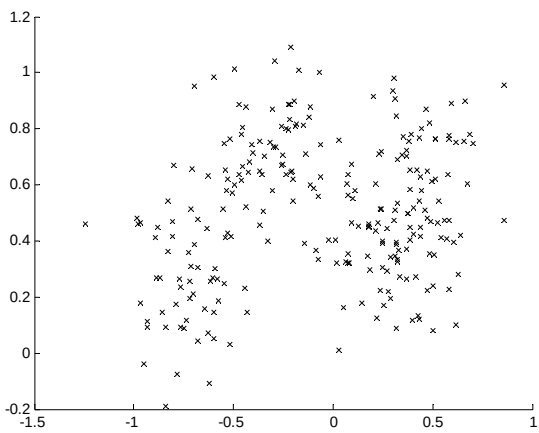
Decrease in distortion cost with iterations



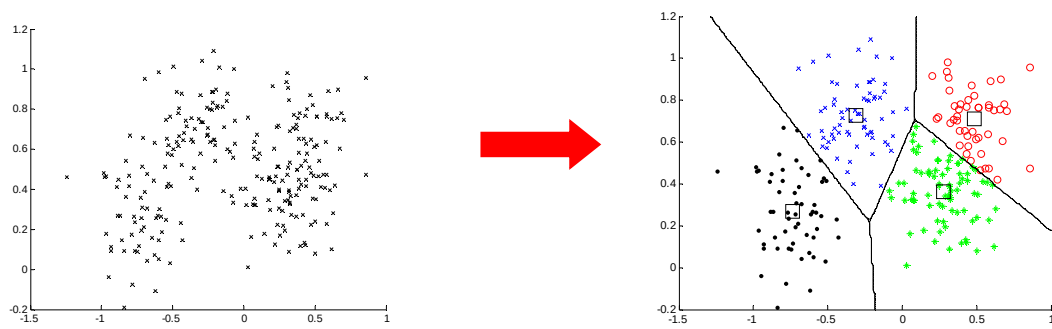
Sensitive to initialization



Sensitive to initialization



Example application 1: vector quantization



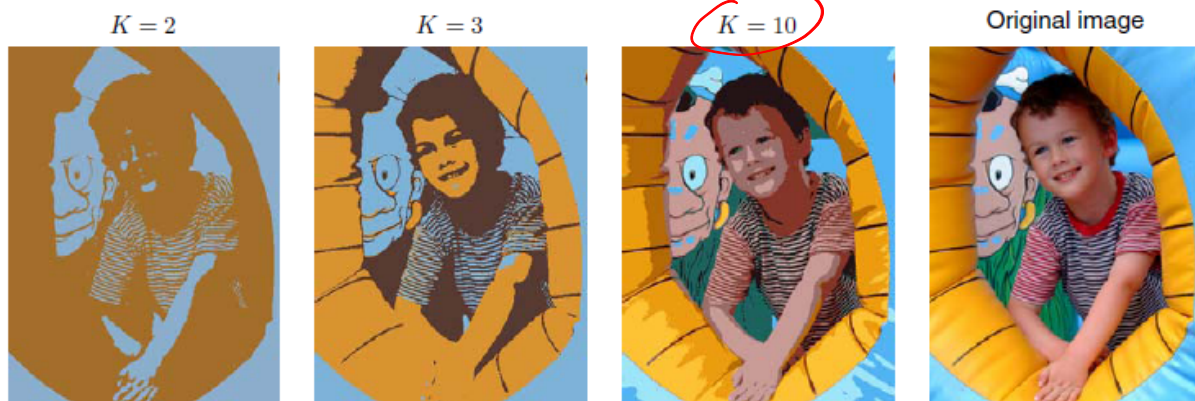
- all vectors in a cluster are considered equivalent
- they can be represented by a single vector – the cluster centre
- applications in compression, segmentation, noise reduction

binary assignment
-1 = 0

Example: image segmentation

- K-means cluster all pixels using their colour vectors (3D)
- assign pixels to their clusters
- colour pixels by their cluster assignment

color compression
true color
24-bit
8-8-8



Example application 2: face clustering

- Determine the principal cast of a feature film
- Approach: view this as a clustering problem on faces

Algorithm outline

1. Detect faces for every fifth frame in the movie
2. Describe the face by a vector of intensities
3. Cluster using a K-means algorithm

Example – “Ground Hog Day” 2000 frames



Subset of detected faces in temporal order



clustering

similarity

C_1

C_2

C_3

C_A

Clusters for $K = 4$



Gaussian Mixture Models

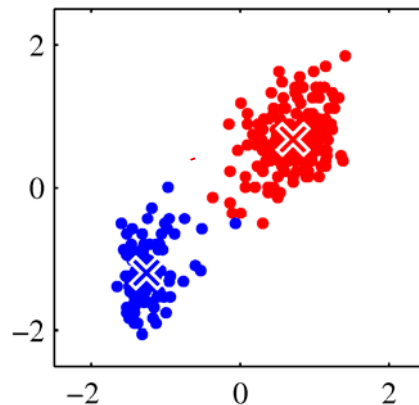
soft clustering

yes/no

probability

Hard vs soft assignments

- In K-means, there is a **hard assignment** of vectors to a cluster
- However, for vectors near the boundary this may be a poor representation
- Instead, can consider a **soft-assignment**, where the strength of the assignment depends on distance



Gaussian Mixture Model (GMM)

Combine simple models
into a complex model:

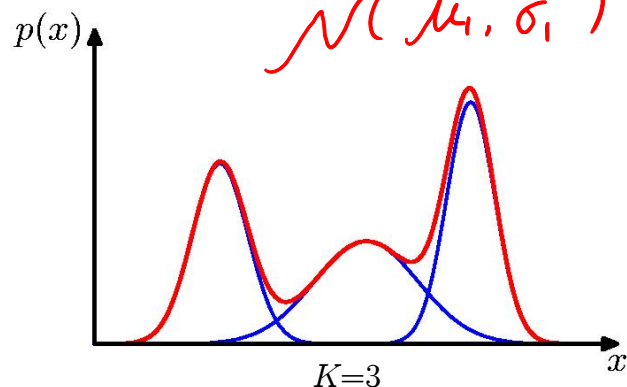
$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$$

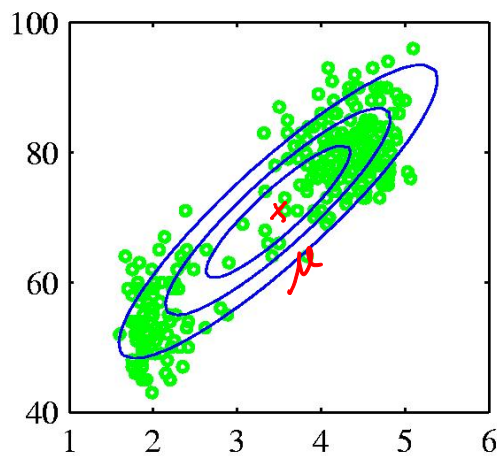
Component

Mixing coefficient

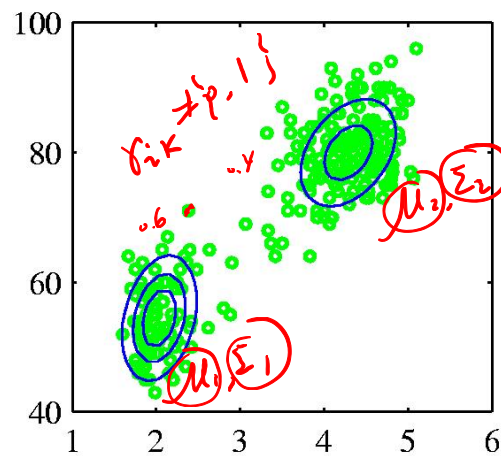
$$\forall k : \pi_k \geq 0$$

$$\sum_{k=1}^K \pi_k = 1$$





Single Gaussian



Mixture of two Gaussians

clustering

Cost function for fitting a GMM

For a point $\mathbf{x}_i$

$$p(\mathbf{x}_i) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \underline{\mu}_k, \Sigma_k)$$

π_k

K clusters
 μ_k, Σ_k

The likelihood of the GMM for N points (assuming independence) is

$$\prod_{i=1}^N p(\mathbf{x}_i) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)$$

and the (negative) log-likelihood is

$$\mathcal{L}(\theta) = - \sum_{i=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)$$

where θ are the parameters we wish to estimate (i.e. μ_k and Σ_k in this case).

elbow

To minimize $\mathcal{L}(\theta)$, differentiate first wrt μ_k

$$\frac{d\mathcal{L}(\theta)}{d\mu_k} = \sum_{i=1}^N \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)} \Sigma_k^{-1} (\mathbf{x}_i - \mu_k)$$

Rearranging

$$\sum_{i=1}^N \gamma_{ik} \mu_k = \sum_{i=1}^N \gamma_{ik} \mathbf{x}_i$$

and hence

$$\mu_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} \mathbf{x}_i$$

where

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)}$$

$$N_k = \sum_{i=1}^N \gamma_{ik}$$

and γ_{ik} are the responsibilities of mixture component k for vector $\mathbf{x}_i$. N_k is the effective number of vectors assigned to component k .

γ_{ik} play a similar role to the assignment variables r_{ik} in K-means, but γ_{ik} is not binary, $0 \leq \gamma_{ik} \leq 1$

γ_{ik} — indicating probabilistic assignment

$\gamma_{ik} \sim 1$

$\mathbf{x}_i \rightarrow \mu_k$

n - point

weighted mean

update mean

Differentiating wrt Σ_k gives

$$\Sigma_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^\top$$

weighted covariance

and wrt π_k (enforcing the constraint that $\sum_k \pi_k = 1$ with a Lagrange multiplier) gives

$$\pi_k = \frac{N_k}{N}$$

which is the average responsibility for the component

Now, ... an algorithm for minimizing the cost function

Expectation Maximization (EM) Algorithm

Step 1 Expectation: Compute responsibilities using current parameters μ_k, Σ_k (assignment)

Calculate expectation

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)}$$

initial $\downarrow$ $\downarrow$ $\downarrow$ *random*
initial μ_k, Σ_k, π_k
 γ_{ik} *random*

Step 2 Maximization: Re-estimate parameters using computed responsibilities

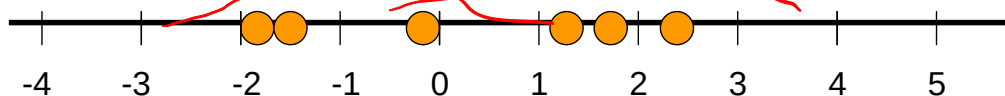
$$\begin{aligned} \mu_k &= \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} \mathbf{x}_i \\ \Sigma_k &= \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^\top \\ \pi_k &= \frac{N_k}{N} \quad \text{where } N_k = \sum_{i=1}^N \gamma_{ik} \end{aligned}$$

Repeat until convergence

Example in 1D

Data:

$$\mathbf{x} = (x_1, x_2, \dots, x_N)$$



OBJECTIVE: Fit mixture of Gaussian model with $K=2$ components

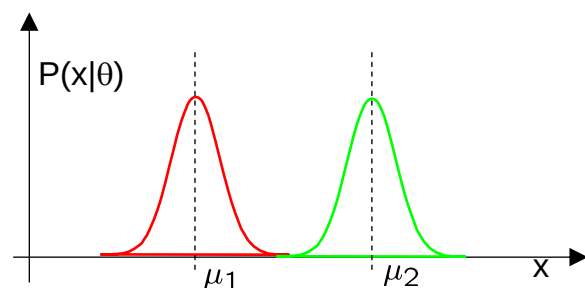
Model:

$$p(x_i | \theta) = \sum_k \pi_k \mathcal{N}(x_i | \mu_k, \sigma_k) \quad \text{where} \quad \sum_{k=1}^K \pi_k = 1$$

Parameters: $\theta = \{\pi, \mu, \sigma\}$

keep π, σ fixed

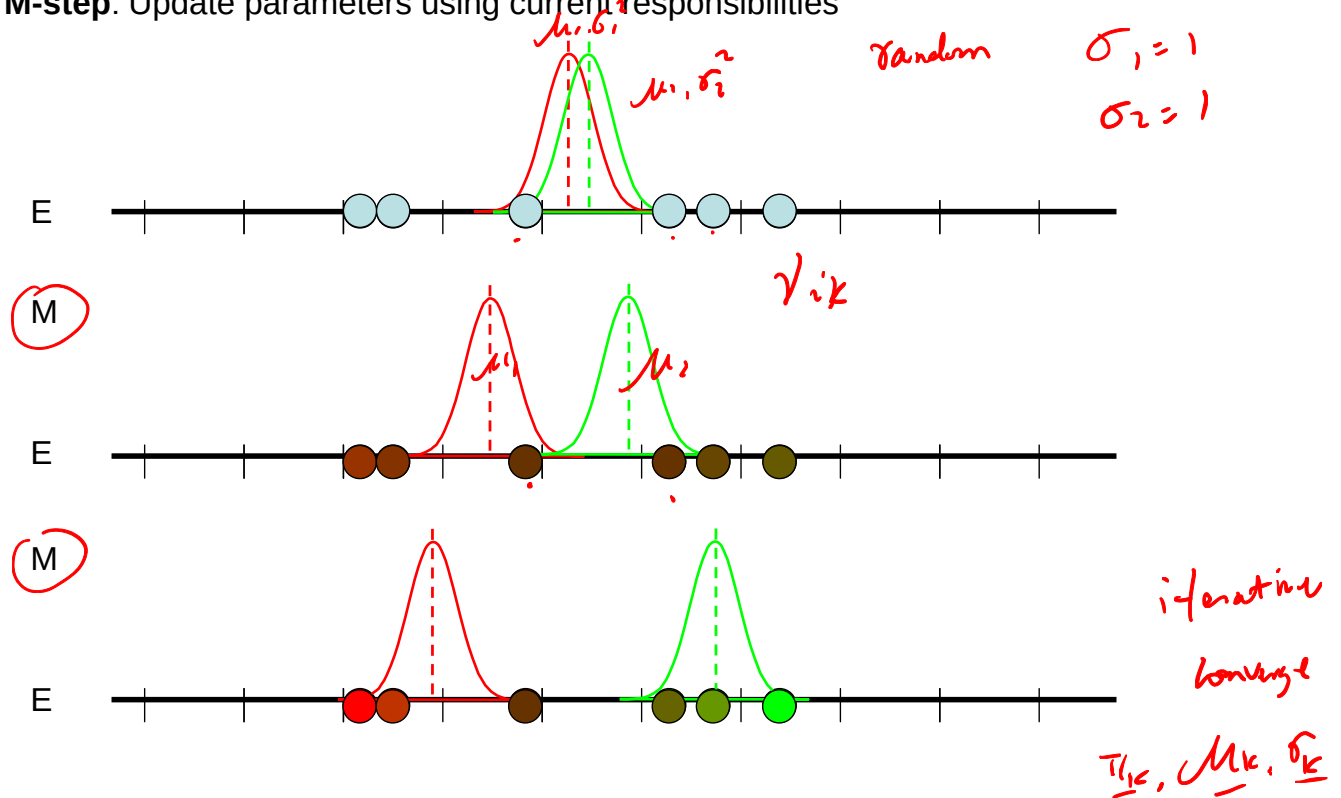
i.e. only estimate μ



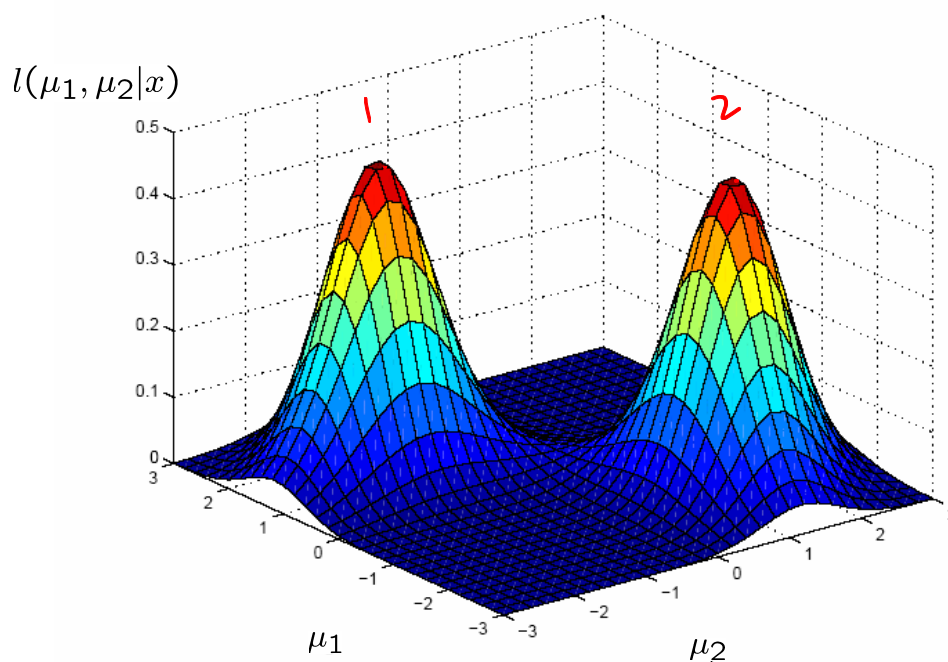
Intuition of EM

E-step: Compute soft assignment of the points, using current parameters

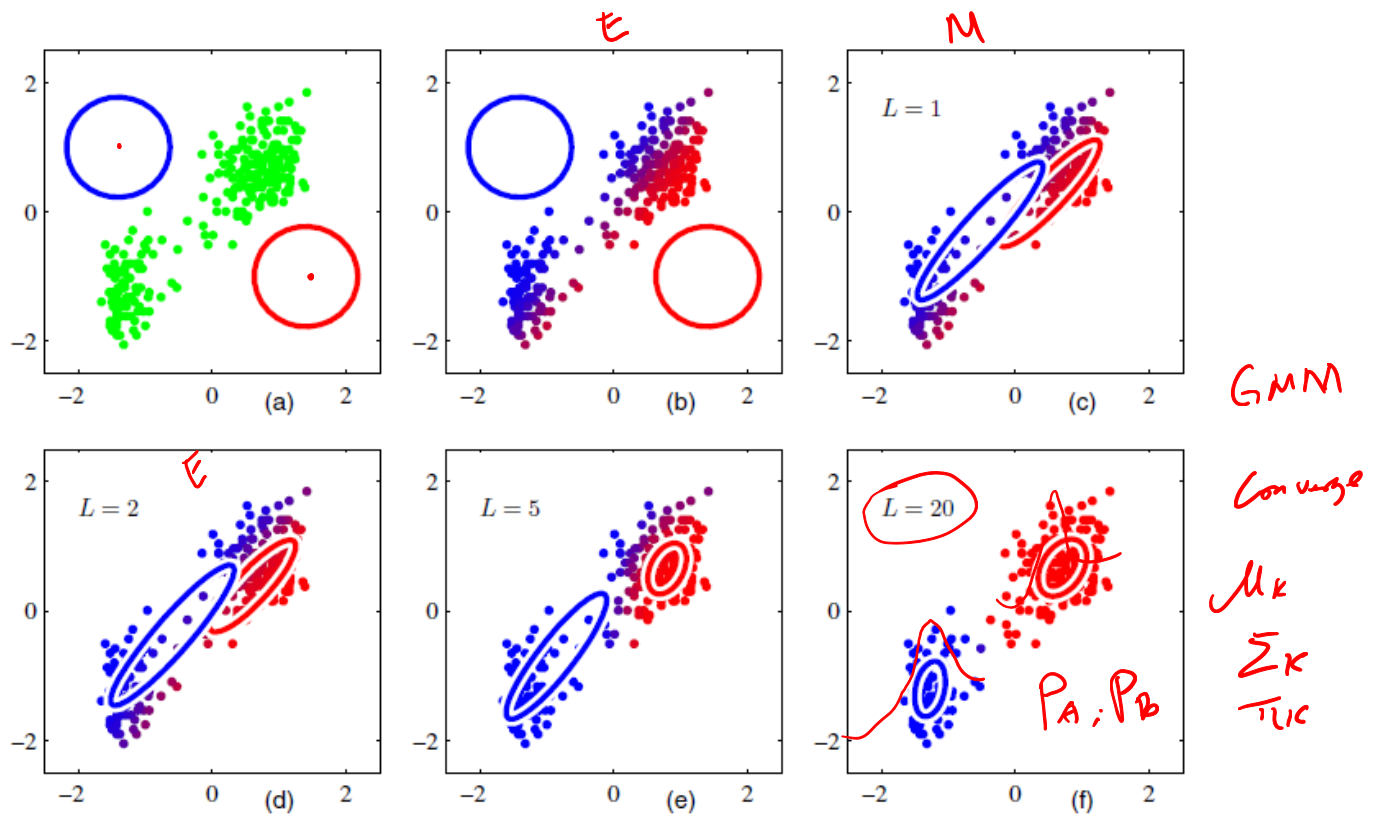
M-step: Update parameters using current responsibilities



2D data example



2D example: fitting means and covariances



Practicalities

- Usually initialize EM algorithm using K-means μ_k, Σ_k

- Choice of K (practical way)

- Can converge to a local rather than global minimum

k-means
— special case of GMM

